

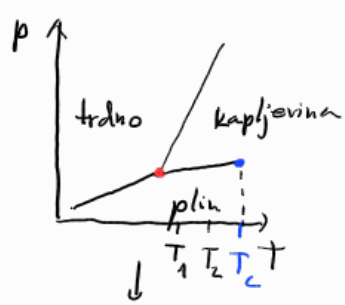
• spremembe

- 1) gostota (kapljevina-plin)
- 2) simetrija (trdnina-kapljevina)
- 3) električne in magnetne lastnosti

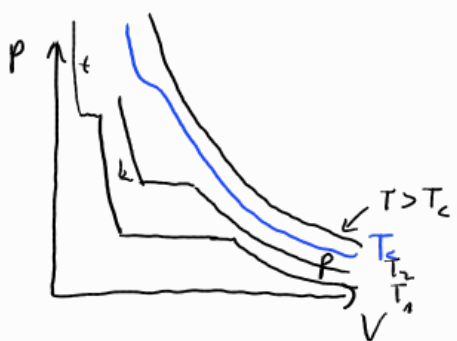
Zgled: led $\rightarrow$ voda $\rightarrow$ para



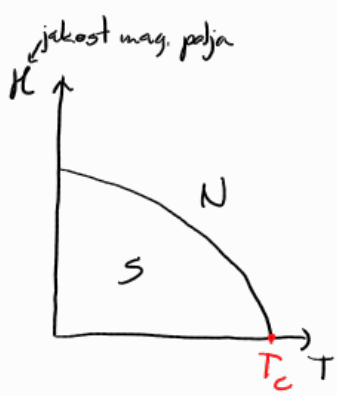
fazni diagram: $T + \text{pospl. zun. sila } (p, \delta l, \dots)$



♦ trojna točka
♦ kritična točka



strmina $p \approx$ strmina t

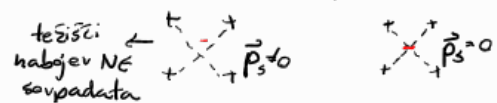


S - supra prevodna faza
N - normalna faza

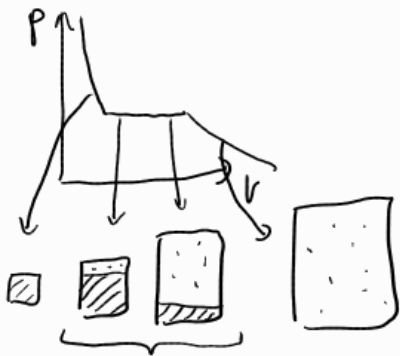
ureditveni parameter (količina, ki je nad kritično točko 0, pod njo pa končna)

- 1) kap.-plin: $\rho_k - \rho_p$
- 2) feromagnet-paramagnet: M_s
 $\downarrow$
 $\vec{M}_s \neq 0$ $\vec{M}_s = 0$
 (spontana magnetizacija)
- 3) ferroelektrik-paraelektrik: $\vec{p}_s$

(feromagnet ima magh. last. tudi v odsotnosti zun. mag. polja)



Ravnovesje faz



ravnovesje
kapljevite in
plinne

- $T_k = T_p$
- $P_k = P_p$
- $\mu_k = \mu_p$ (kemijski potencial)

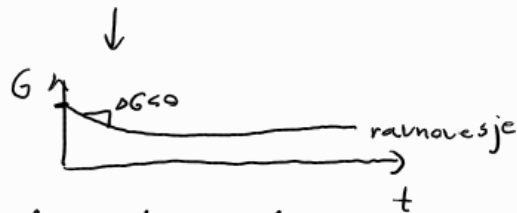
Kemijski potencial

$$\mu_i = \left(\frac{\partial G}{\partial m_i} \right)_{T, m_{j \neq i}, p} \quad \leftarrow \text{prosta entalpija}$$

$$\mu_k = \left(\frac{\partial G}{\partial m_k} \right)_{T, m_p, p}$$

$$\mu_p = \left(\frac{\partial G}{\partial m_p} \right)_{T, m_k, p}$$

$$dG = -SdT + Vdp + \underbrace{\mu_k}_{\text{prosta entalpija}} dm_k + \underbrace{\mu_p}_{\text{prosta entalpija}} dm_p$$



$$dG = \mu_k dm_k + \mu_p dm_p \leq 0$$

$$m_k + m_p = \text{konst.}$$

$$dm_k + dm_p = 0 \rightarrow dm_k = -dm_p$$

$$dG = (\mu_k - \mu_p) dm_k < 0$$

$$\left. \begin{array}{l} 1) \mu_k > \mu_p \Rightarrow dm_k < 0 \\ 2) \mu_k < \mu_p \Rightarrow dm_k > 0 \end{array} \right\} \begin{array}{l} \text{snov se seli v fazo z} \\ \text{nižjim } \mu \end{array}$$

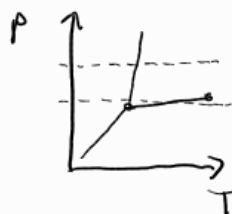
Zvezen, nezvezen fazni prehod (1. vrste) (2. vrste)

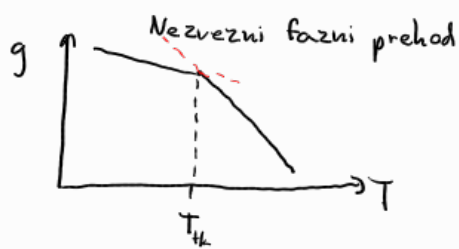
prosta entalpija: p, T

$$\frac{dG}{m} = dg = s dT + \frac{V}{m} dp = -s dT + \frac{dp}{\rho}$$

$$\rightarrow dg = -s dT$$

ponavadi pri eksperimentih: $p = \text{konst.}$

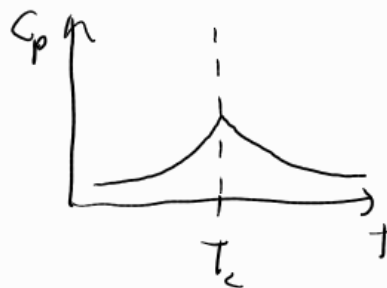
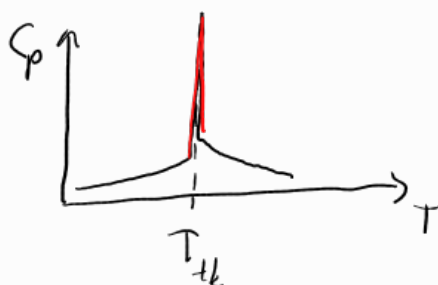
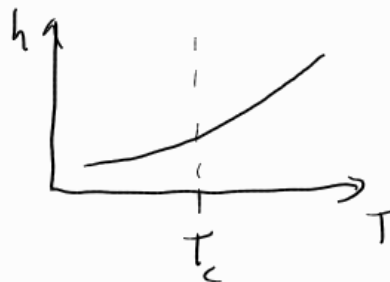
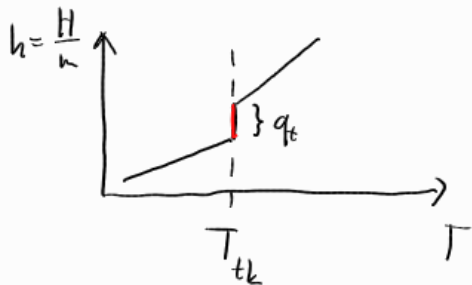




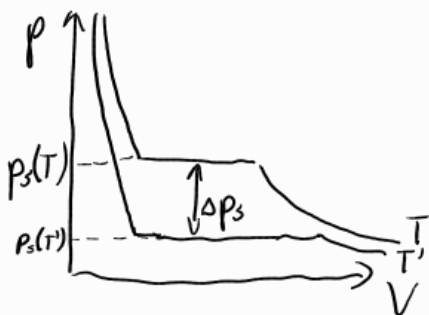
$$q_t = T \Delta s = T(s_k - s_l) = -T \left[\left(\frac{\partial g}{\partial T} \right)_k - \left(\frac{\partial g}{\partial T} \right)_l \right]$$



$$q_t = 0$$



Clusius-Clapeyronova enačba



p_s ... nasičeni parni tlak

$$\left(\frac{\partial p}{\partial T} \right)_V = \left(\frac{\partial S}{\partial V} \right)_p$$

↓

$$\Delta S = \frac{Q}{T} = \frac{mq_i}{T}$$

$$\frac{dp_s}{dT} = \frac{\Delta S}{\Delta V} = \frac{s_p - s_k}{V_p - V_k}$$

$$\frac{dp_s}{dT} = \frac{mq_i}{T(V_p - V_k)}$$

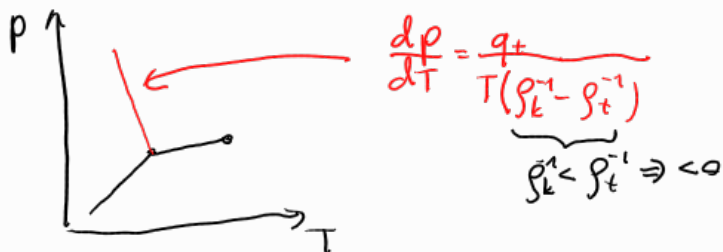
$$\boxed{\frac{dp_s}{dT} = \frac{q_i}{T(p_p' - p_k')}} \quad \text{where } p_p' = p_p(T), p_k' = p_k(T)$$

$$\text{če } T \ll T_c \rightarrow V_p \gg V_k, \rho_p^{-1} = \frac{V_p}{m} = \frac{RT}{M p_s}$$

$$\rightarrow \frac{dp_s}{dT} = \frac{q_i M p_s}{R T^2}, \quad \frac{dp_s}{p_s} = \frac{M q_i}{R} \frac{dT}{T^2}$$

$$\rightarrow \boxed{p_s(T) = p_s(T') \exp\left(-\frac{M q_i}{R} \left(\frac{1}{T} - \frac{1}{T'}\right)\right)}$$

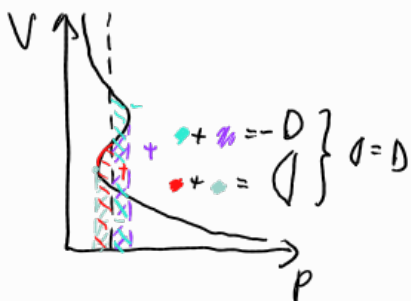
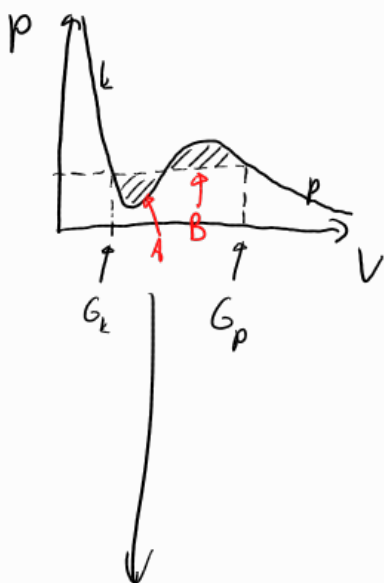
Fazni diagram vode



Maxwellovo pravilo

Van der Waalsova enačba stanja:

$$\left(p + \frac{a}{V_m^2}\right)(V_m - b) = RT$$



ravnovesje faz
 $G_p = G_k$ (ekvivalentno $\mu_p = \mu_k$ za $m = \text{konst.}$)

$$G_p = G_k + \int_k^p dG$$

$$\hookrightarrow dG = -SdT + Vdp$$

||
po izotermi

$$G_p = G_k + \int V dp$$

Maxwellovo pravilo: za ravnovesje faz velja $\int V dp = 0$
 aka ploščini A in B sta enaki

